

Hanna Abi Akl

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SUMMARY

I am a Research Scientist in Artificial Intelligence. My research interests include neuro-symbolic artificial intelligence, knowledge representation, knowledge graphs, ontologies, natural language processing and the semantic web. I am currently working on neuro-symbolic and knowledge-based approaches for reasoning abilities in Large Language Models. I also teach at Data ScienceTech Institute and Sciences Po.

PROFESSIONAL EXPERIENCE

- **Data ScienceTech Institute School of Engineering**  02/2023 - Current
Vice-Dean and Chief Academic Officer Paris, France
 - Teaching activities: Clean IT, Software Engineering, Machine Learning Labs, Artificial Neural Networks, Deep Learning, Amazon Web Services, Data Wrangling with SQL
 - Preparation of Faculty Seminars and scientific programme reviews
 - Improvement of programme structure and evaluation (Bachelors and Masters level)
 - Responsible for faculty recruitment interviews
- **INRIA**  12/2025 - Current
PhD Student Nice, France
 - Part of WIMMICS, 3IA, CNRS, I3S and Université Côte d'Azur
 - Topic: Neuro-symbolic and knowledge-based approaches for reasoning in Large Language Models
 - Supervision: Fabien Gandon, Catherine Faron, Pierre Monnin
 - Keywords: Logical Reasoning, Neuro-symbolic Artificial Intelligence, Knowledge Graphs, Ontologies, Natural Language Processing, Semantic Web
 - Scientific Events: ESSAI 2025, Reasoning Web 2025, SeReCo 2026, RuleML+RR 2026
- **SciencesPo Paris School of International Affairs**  08/2024 - Current
Adjunct Instructor Paris, France
 - Teaching activities: Programming, Databases, and Machine Learning Foundations for Decision-Makers, Cloud Computing
- **INRIA**  02/2023 - 11/2025
External Scientific Collaborator Nice, France
 - Member of the WIMMICS team
 - PhD thesis supervision on triplet relation extraction using NLP and Semantic Web methodologies
 - Testing and validation of the KGStat tutorial project based on Machine Learning techniques using Semantic Web data
- **Yseop**  04/2019 - 02/2023
Data Scientist and Machine Learning Engineer Paris, France
 - Identification and prototyping of artificial intelligence use cases using machine and deep learning techniques
 - Design and integration of artificial intelligence tools in existing product from text corpora, APIs, business ontologies and knowledge graphs
- **Data ScienceTech Institute School of Engineering**  09/2021 - 02/2023
Course Instructor Paris, France
 - Teaching activities: Clean IT, Software Engineering, Machine Learning Labs
- **Data ScienceTech Institute School of Engineering**  01/2019 - 04/2019
Private Tutor Paris, France
 - Preparing students for the "AWS Certified Solutions Architect - Associate" certification exam
- **Data Consult**  06/2015 - 08/2018
Data Integration Engineer Beirut, Lebanon
 - Configuration, maintenance and customization of cloud applications
 - High-level analysis and smart data reporting through API and chatbot software integration
 - Implementation of first Smart City solution in Lebanon
 - Presentation of data augmentation platform at the Mobile World Congress (MWC)
- **Touch**  07/2014 - 08/2014
Site Implementation and Maintenance (SI&M) Engineering Intern Beirut, Lebanon

- Theoretical research on the development and advancement of 4G technology
- **American University of Beirut**  06/2012 - 12/2013
Beirut, Lebanon
Research Assistant
 - Benchmarking and optimization of Alaoui filter and integrator model parameters in Matlab
- **American University of Beirut**  09/2011 - 09/2015
Beirut, Lebanon
Private Tutor
 - Teaching middle school and high school subjects

EDUCATION

- **Université Côte d'Azur** 12/2025 - Current
Nice, France
PhD, Computer Science
 - Topic: Neuro-symbolic and knowledge-based approaches for reasoning in Large Language Models
 - Supervision: Fabien Gandon, Catherine Faron, Pierre Monnin
- **Quantic School of Business and Technology** 12/2024 - 06/2026
Washington, DC, United States of America
Executive MBA
 - GPA: 4.0/4.0
 - Awarded the France Leadership scholarship
- **Université Côte d'Azur** 12/2023 - 07/2024
Nice, France
Master, Computer Science
 - GPA: 4.0/4.0
 - Awarded full completion by the Validation d'Acquis par Expérience (VAE)
- **Data ScienceTech Institute School of Engineering** 10/2018 - 09/2019
Paris, France
M.Sc., Data Science and Artificial Intelligence
 - GPA: 4.0/4.0
 - Graduated Summa Cum Laude
 - Awarded Second prize at HackBordeaux 2019 Capgemini challenge for the FuelMi application
- **American University of Beirut** 09/2011 - 05/2015
Beirut, Lebanon
B.E., Computer and Communications Engineering
 - Dean's Honor List: Spring Semester 2015
 - Awarded Second prize in the National Instruments elevator design challenge
 - Recipient of the Philippe Jabre Association scholarship for excellence
- **Collège Notre Dame de Jamhour** 09/1998 - 06/2011
Beirut, Lebanon
High School Diploma, General Sciences
 - Ranked in top 10% of cohort (300 students)
 - High Distinction in the French Baccalaureate
 - High Distinction in the Lebanese Baccalaureate
 - Recipient of the Carlos Ghosn scholarship for excellence

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, O=OTHER, P=PATENT, S=SOFTWARE, T=THESIS

- [O.3] **Hanna Abi Akl** (2026). **Logic in the Era of Artificial Intelligence: Common Logic Grammar Construction Framework**. In *HAL Open Science*.
- [O.2] **Hanna Abi Akl** (2026). **Logic in the Era of Artificial Intelligence: A Case Study on Syllogisms**. In *HAL Open Science*.
- [C.21] **Hanna Abi Akl**, Fabien Gandon, Catherine Faron, Pierre Monnin (2026). **Are you Talking Logic to Me? Assessing Language Models Syllogistic Reasoning Capabilities**. In *the International Joint Conference on Rules and Reasoning (RuleML+RR) 2026*.
- [J.8] Ayman Ali Sharara, **Hanna Abi Akl** (2026). **IdiomX A Multilingual Benchmark for Idiom Understanding, Retrieval, and Interpretation**. In *arXiv preprint arXiv:2606.02584*.
- [S.1] **Hanna Abi Akl** (2026). **CLGC**.
- [C.20] **Hanna Abi Akl**, Fabien Gandon, Catherine Faron, Pierre Monnin (2026). **SEF-CLGC at SemEval-2026 Task 11: Logical Notation Impact on Language Model Performance**. In *Proceedings of The 20th International Workshop on Semantic Evaluation (SemEval-2026) co-located with 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*.
- [J.7] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2026). **SHACL-Guided Small Language Models for RDF Knowledge Graph Population: Semantic Evaluation, Cross-Domain Generalisation, and Long-Tail Mitigation**. In *Semantic Web journal*.

- [C.19] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2026). **Généraliser l'adaptation de modèles de langue frugaux pour l'extraction de motifs RDF à partir de texte, à des relations de type Datatype et Object property**. In *EGC 2026 - 26^{ème} conférence francophone sur l'extraction et la gestion des connaissances*.
- [J.6] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2025). **A systematic review of relation extraction task since the emergence of Transformers**. In *arXiv preprint arXiv:2511.03610*.
- [J.5] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2025). **Overcoming the Generalization Limits of SLM Finetuning for Shape-Based Extraction of Datatype and Object Properties**. In *Proceedings of the 13th Knowledge Capture Conference 2025*.
- [C.18] **Hanna Abi Akl** (2025). **Investigating Language Model Capabilities to Represent and Process Formal Knowledge: A Preliminary Study to Assist Ontology Engineering**. In *the International Joint Conference on Rules and Reasoning (RuleML+RR) 2025 Doctoral Consortium*.
- [C.17] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2025). **Kastor: Fine-tuned Small Language Models for Shape-based Active Relation Extraction**. In *European Semantic Web Conference*, pp. 94-115. Springer, Cham. DOI: 10.1007/978-3-031-94575-5_6.
- [J.4] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2025). **Extensive Benchmark of Frugal Encoder-Decoder Language Models for Datatype Properties Extraction and RDF Knowledge Graph Generation**. In *Semantic Web*.
- [P.3] **Hanna Abi Akl**, Hugues de Mazancourt, Dominique Mariko (2024). **Ontology mapping and adaptation systems and methods**. U.S. Patent Office, Patent No. 12,106,049. Publication Date: 01 October 2024.
- [C.16] **Hanna Abi Akl**, et al (2024). **Knowledge Graphs can play together: Addressing knowledge graph alignment from ontologies in the biomedical domain**. In *Proceedings of the 16th International Conference on Knowledge Discovery and Information Retrieval (KDIR 2024)*, pp. 416-423. DOI: 10.5220/0013015800003838.
- [C.15] **Hanna Abi Akl** (2024). **Project SHADOW: Symbolic Higher-order Associative Deductive reasoning On Wikidata using LM probing**. In *Proceedings of the 10th International Conference on Natural Language Computing (NATL 2024)*.
- [C.14] **Hanna Abi Akl** (2024). **DSTI at LLMs4OL 2024 Task A: Intrinsic versus extrinsic knowledge for type classification**. In *Open Conference Proceedings*, pp. 93-101. DOI: <https://doi.org/10.52825/ocp.v4i.2492>.
- [C.13] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2024). **Learning Pattern-Based Extractors from Natural Language and Knowledge Graphs Applying Large Language Models to Wikipedia & the Linked Open Data**. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 23411-23412. DOI: <https://doi.org/10.1609/aaai.v38i21.30406>.
- [C.12] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2024). **12 shades of RDF: Impact of Syntaxes on Data Extraction with Language Models**. In *European Semantic Web Conference*, pp. 81-91. Springer, Cham. DOI: https://doi.org/10.1007/978-3-031-78952-6_8.
- [C.11] **Hanna Abi Akl** (2024). **NeSy is alive and well: A LLM-driven symbolic approach for better code comment data generation and classification**. In *Joint Proceedings of the ESWC 2024 Workshops and Tutorials co-located with 21th European Semantic Web Conference (ESWC 2024)*.
- [C.10] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2024). **Well-written knowledge graphs: most effective RDF syntaxes for triple linearization in end-to-end extraction of relations from texts**. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 23631-23632. DOI: <https://doi.org/10.1609/aaai.v38i21.30502>.
- [O.1] **Hanna Abi Akl** (2023). **The three pillars of AI: Symbols, Neurons and Graphs**. In *The AI Journal*.
- [C.9] **Hanna Abi Akl** (2023). **PSYCHIC: A Neuro-Symbolic Framework for Knowledge Graph Question-Answering Grounding**. In *Joint Proceedings of Scholarly QALD 2023 and SemREC 2023 co-located with 22nd International Semantic Web Conference (ISWC 2023)*.
- [C.8] **Hanna Abi Akl** (2023). **A ML-LLM pairing for better code comment classification**. In *FIRE (Forum for Information Retrieval Evaluation)*.
- [C.7] Célian Ringwald, Fabien Gandon, Catherine Faron, Franck Michel, **Hanna Abi Akl** (2023). **Learning RDF pattern extractors from natural language and knowledge graphs-application to Wikipedia and the LOD**. In *Proceedings of the AAAI Conference on Artificial Intelligence*. DOI: <https://doi.org/10.1609/aaai.v38i21.30406>.
- [P.2] **Hanna Abi Akl**, Hugues de Mazancourt, Dominique Mariko, Valentin Pierart (2023). **Systems and methods for measuring automatability of report generation using a natural language generation system**. U.S. Patent Office, Patent No. 11,734,517. Publication Date: 22 August 2023.
- [C.6] **Hanna Abi Akl** (2023). **The Path to Autonomous Learners**. In *Science and Information Conference*, pp. 808-830. DOI: https://doi.org/10.1007/978-3-031-37717-4_52.

- [P.1] Dominique Mariko, **Hanna Abi Akl**, Hugues de Mazancourt (2022). **Techniques for generating natural language text customized to linguistic preferences of a user**. U.S. Patent Office, Patent No. 11,501,088. Publication Date: 15 November 2022.
- [J.3] Rami Bou Khalil, Ramez Dagher, Myriam Zarzour, Ghassan Sleilaty, **Hanna Abi Akl**, et al (2022). **The impact of lockdown and other stressors during the COVID-19 pandemic on depression and anxiety in a Lebanese opportunistic sample: an online cross-sectional survey**. In *Current Psychology*, pp. 1-11. DOI: <https://doi.org/10.1007/s12144-021-02644-0>.
- [C.5] **Hanna Abi Akl**, et al (2021). **Yseop at FinSim-3 Shared Task 2021: Specializing Financial Domain Learning with Phrase Representations**. In *Proceedings of the Third Workshop on Financial Technology and Natural Language Processing*, pp. 52–57.
- [C.4] Anubhav Gupta, **Hanna Abi Akl**, et al (2021). **Not all titles are created equal: Financial document structure extraction shared task**. In *Proceedings of the 3rd Financial Narrative Processing Workshop*, pp. 86-88.
- [C.3] **Hanna Abi Akl**, et al (2020). **Yseop at SemEval-2020 Task 5: Cascaded BERT Language Model for Counterfactual Statement Analysis**. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, pp. 468–478.
- [J.2] **Hanna Abi Akl**, et al (2020). **SemEval-2020 Task 5: Detecting Counterfactuals by Disambiguation**. In *arXiv preprint arXiv:2005.08519*.
- [J.1] Dominique Mariko, Estelle Labidurie, Yagmur Ozturk, **Hanna Abi Akl**, et al (2020). **Data Processing and Annotation Schemes for FinCausal Shared Task**. In *arXiv preprint arXiv:2012.02498*.
- [C.2] Dominique Mariko, **Hanna Abi Akl**, et al (2020). **Financial Document Causality Detection Shared Task (FinCausal 2020)**. In *Proceedings of the 1st Joint Workshop on Financial Narrative Processing and MultiLing Financial Summarisation*, pp. 23-32.
- [C.1] **Hanna Abi Akl**, Anubhav Gupta, Dominique Mariko (2019). **Fintoc-2019 shared task: Finding title in text blocks**. In *Proceedings of the Second Financial Narrative Processing Workshop (FNP 2019)*, pp. 58-62.

PROJECTS

D=DATA, P=PROJECT, S=SOFTWARE

- **[D.7] P-FOLIO-KR** 2026
Tools: Python, Hugging Face [🔗]
 - Implemented scripts for first-order logic transformation to knowledge representation languages
 - Created custom formal knowledge representation languages
 - Publicly released on Hugging Face
- **[S.12] CLGC** 2026
Tools: Python, GitHub [🔗]
 - Python package to translate first-order logic into different knowledge representation languages using BNF grammar and Abstract Syntax Trees
 - Support for multiple knowledge representation languages
 - Categorizes syllogisms by type
- **[S.11] LOGIC-NL-CLIF-Flan-T5-Large** 2026
Tools: Python, Deep Learning, Hugging Face [🔗]
 - Logic language model fine-tuned to classify syllogism veracity in Common Logic Interchange Format
 - Trained on SemEval-2026 Task 11 dataset
 - Publicly released on Hugging Face
- **[S.10] LOGIC-NL-Flan-T5-Large** 2026
Tools: Python, Deep Learning, Hugging Face [🔗]
 - Logic language model fine-tuned to classify syllogism veracity in natural language
 - Trained on SemEval-2026 Task 11 dataset
 - Publicly released on Hugging Face
- **[D.6] FOLIO-KR** 2026
Tools: Python, Hugging Face [🔗]
 - Implemented scripts for first-order logic transformation to knowledge representation languages
 - Created custom formal knowledge representation languages
 - Publicly released on Hugging Face
- **[S.9] SHADOW: LM-probing model for triple completion on Wikidata** 2024
Tools: Python, SPARQL, Wikidata, Knowledge Graphs, Deep Learning, Hugging Face [🔗]
 - Fine-tuned model on data
 - Publicly released on Hugging Face

• [S.8] FLAN-T5 small-WordNet: WordNet term classification with Wikidata semantic information <i>Tools: Python, SPARQL, Wikidata, Knowledge Graphs, Deep Learning, Hugging Face</i> ◦ Extracted Wikidata semantic information ◦ Created semantic embedding vectors ◦ Fine-tuned model on data ◦ Publicly released on Hugging Face	2024 
• [S.7] FLAN-T5 small-GeoNames: GeoNames term classification with Wikidata semantic information <i>Tools: Python, SPARQL, Wikidata, Knowledge Graphs, Deep Learning, Hugging Face</i> ◦ Extracted Wikidata semantic information ◦ Created semantic embedding vectors ◦ Fine-tuned model on data ◦ Publicly released on Hugging Face	2024 
• [D.5] WordNet Semantic Primes: Enriching WordNet data with Wikidata <i>Tools: Python, SPARQL, MongoDB, Wikidata, Hugging Face</i> ◦ Created scripts to query Wikidata Knowledge Graph information ◦ Vectorized Knowledge Graph information ◦ Configured MongoDB to store and index vectors ◦ Publicly released on Hugging Face	2024 
• [D.4] GeoNames Semantic Primes: Enriching GeoNames data with Wikidata <i>Tools: Python, SPARQL, MongoDB, Wikidata, Hugging Face</i> ◦ Created scripts to query Wikidata Knowledge Graph information ◦ Vectorized Knowledge Graph information ◦ Configured MongoDB to store and index vectors ◦ Publicly released on Hugging Face	2024 
• [P.5] KGstat: Knowledge Graph querying and machine learning modeling on data <i>Tools: Python, R, SPARQL, Knowledge Graphs, Machine Learning</i> ◦ Applied statistical analysis for classification and regression cases on Knowledge Graph data ◦ Proposed machine learning models for improving classification and regression performance ◦ Evaluated used metrics for classification and regression	2023 
• [S.6] PSYCHIC: Question-answering neuro-symbolic model on knowledge graphs with SPARQL <i>Tools: Python, SPARQL, Knowledge Graphs, Deep Learning, Hugging Face</i> ◦ Fine-tuned model on data ◦ Publicly released on Hugging Face	2023 
• [S.5] Text Smoothing: Generative model for financial text rewriting <i>Tools: Python, Deep Learning, Hugging Face</i> ◦ Created scripts for scraping and cleaning data pipeline ◦ Tested model zero-shot configurations ◦ Publicly released on Hugging Face	2022 
• [S.4] Complete FNP T5 Data2Text: Complete T5-base data to text model specialized for finance NLG <i>Tools: Python, Deep Learning, Hugging Face</i> ◦ Created scripts for scraping and cleaning data pipeline ◦ Re-trained model on data ◦ Publicly released on Hugging Face	2022 
• [S.3] Simple FNP T5 Data2Text: Simple T5-base data to text model specialized for finance NLG <i>Tools: Python, Deep Learning, Hugging Face</i> ◦ Created scripts for scraping and cleaning data pipeline ◦ Fine-tuned model on data ◦ Publicly released on Hugging Face	2022 
• [D.3] FNP 2022 FinCausal: Data set for the FinCausal 2022 shared task <i>Tools: Python</i> ◦ Performed data curation ◦ Participated in inter-annotator agreement sessions ◦ Created baseline models and scoring scripts	2022 
• [S.2] FinISH: Financial term classification model for domain ontology classes <i>Tools: Python, Ontologies, Deep Learning, Hugging Face</i>	2021 

- Created scripts for scraping and cleaning FIBO ontology
- Augmented financial ontology terms with definitions
- Fine-tuned model on data
- Publicly released on Hugging Face
- **[S.1] Financial Relation Extraction: Classification model for type of relations in financial sentences** 2021
Tools: Python, Deep Learning, Hugging Face 
- Created scripts for scraping and cleaning financial data pipeline
- Extracted and normalized financial relations
- Fine-tuned model on data
- Publicly released on Hugging Face
- **[D.2] FNP 2021 FinCausal: Data set for the FinCausal 2021 shared task** 2021
Tools: Python 
- Performed data curation
- Participated in inter-annotator agreement sessions
- Created baseline models and scoring scripts
- **[D.1] FNP 2020 FinCausal: Data set for the FinCausal 2020 shared task** 2020
Tools: Python 
- Performed data curation
- Participated in inter-annotator agreement sessions
- Created baseline models and scoring scripts
- **[P.4] FuelMi: The pet that takes care of you** 2019
Tools: Python 
- Designed application concept and intent
- Implemented Python back-end functionalities for health and activity management
- Pitched in front of live jury
- **[P.3] Lebanon Smart City Experience: First smart city in Lebanon at BeitMisk** 2017
Tools: Air Quality Sensors, Node-Red, PHP, MySQL, HTML5, MQTT Protocol, Server-Side JavaScript 
- Maintained and queried sensor log data in MySQL database
- Created PHP query scripts to provide data insights for monitoring dashboard
- Updated HTML5 dashboard with insights on consumption tracking
- Developed custom Facebook chatbot with Node-Red and Watson NLP toolkit
- **[P.2] Ticketing System: Customizing Service Delivery** 2016
Tools: Oracle RightNow Service Cloud, PHP, HTML5, CSS, JavaScript, ROQL, Google Scripts, Google Sheets 
- Implemented and configured ticketing system
- Managed user access and administration
- Created custom workflows for internal use
- Created and enabled custom views and reports
- Managed and customized web portal with HTML5, CSS and JavaScript
- Created custom project management time sheet with Google Sheets back-end and Google Scripts
- Maintained PHP scripts for updated ROQL real-time queries and ticket views
- **[P.1] Sales CRM System: Improving Customer Relationship Management Experience** 2016
Tools: Oracle Sales Cloud, Groovy scripting, Microsoft Dynamics 
- Performed migration from Microsoft Dynamics to Oracle Sales Cloud
- Created custom views for users
- Managed user access and adlinistration
- Setup Single sign-on (SSO) with LDAP between Microsoft Active Directory and Oracle
- Created custom workflows and scripts for internal behavior

SUPERVISION

M=M.Sc., P=PhD

- **[M.4] Ayman Sharara** 06/2023 - 04/2026 
Data ScienceTech Institute School of Engineering
 - Supervised "Deep Learning" research project: IdiomX: English-Arabic Idiom Understanding Dataset
 - Dataset published on Hugging Face and Zenodo
 - Research paper published on ArXiv
- **[M.3] Claire Davat, Albin Brogialdi, Marius Akre, Florian Irrien, Luc Sagnes** 11/2024 - 04/2025 
Data ScienceTech Institute School of Engineering
 - Supervised DIRISI Hackathon project: "L'Intelligence Artificielle au cœur des réseaux"
 - Ranked Second in the challenge
- **[M.2] Jannic Cutura** 06/2023 - 11/2023 
Data ScienceTech Institute School of Engineering
 - Supervised "Deep Learning" research project: Breaking the U: Asymmetric U-Net for Object Recognition in Muon Tomography
 - Ranked Fourth in the IEEE BigData 2023 Cup: Object Recognition with Muon Tomography using Cosmic Rays challenge
 - Paper publication in IEEE Conference Proceedings
- **[P.1] Célian Ringwald** 12/2022 - Current 
Université Côte d'Azur, Inria, CNRS, I3S
 - Supervised PhD thesis "Learning RDF pattern extractors for a language from dual Wikipedia/LOD databases"
- **[M.1] Jannic Cutura** 12/2022 - 04/2023 
Data ScienceTech Institute School of Engineering
 - Supervised "Python Machine Learning Labs" research project: Web Intelligence - Deduplication challenge
 - Ranked First (reproducibility) and Second (accuracy) in the EuropeanStatisticsAwards challenge
 - Prize money: 14,000 euros

SKILLS

- **Programming Languages:** Python, C, C++, Java, JavaScript, PHP, HTML5, CSS3
- **Web Technologies:** RDF, RDFS, OWL, SPARQL
- **Database Systems:** Microsoft SQL Server, MySQL, PostgreSQL, MongoDB, Neo4j
- **Data Science & Machine Learning:** Classification, Regression, Clustering, Reinforcement Learning
- **Cloud Technologies:** Amazon Web Services
- **DevOps & Version Control:** Git, SVN
- **Specialized Area:** Natural Language Processing, Knowledge Graphs, Computational Linguistics
- **Mathematical & Statistical Tools:** Matlab, Latex
- **Other Tools & Technologies:** Docker, Windows, Linux (bash scripting)
- **Project Management:** Agile methods (Scrum), Test-Driven Development

HONORS AND AWARDS

- **Second Place in the LLMs4OL challenge** 2024 
International Semantic Web Conference (ISWC)
 - Second in F1 score (97.16%) in SubTask A.1 (FS) - Term Typing - Wordnet
- **Third Place in the Scholarly-QALD challenge** 2023 
International Semantic Web Conference (ISWC)
 - Third in F1 score (71.00%) in DBLP-QuAD Entity Linking
- **Second Place in the IRSE shared task** 2023 
FIRE: Forum for Information Retrieval Evaluation
 - Second in F1 score increase (+1.5%)
- **Company Core Values Award Q3 Trust** 2022 
Yseop
 - Chosen by colleagues (80 employees)
- **Second place in the FinSim-3 shared task** 2021 
FinNLP
 - Second in Mean Rank (1.141) and Accuracy (91.70%) scores
- **Company Core Values Award Q1 Innovation** 2021 
Yseop

- Chosen by colleagues (80 employees)
- **Second Place in the HackBordeaux Capgemini challenge** 2019
Enseirb - Matmeca 
- Second for the FuelMi health application
- **Second Place in the National Instruments Elevator Competition** 2011
American University of Beirut 
- Second in the Elevator Competition design challenge
- **Finalist in the International College School Poetry Competition** 2010
International College 
- Finalist among 300 national participants
- **Recipient of the "Philippe Jabre Foundation Scholarship for Excelling Students"** 2011-2015
American University of Beirut 
- Recipient for academic excellence
- **Recipient of the "Carlos Ghosn Foundation Scholarship for Excelling Students"** 1997-2011
Collège Notre Dame de Jamhour 
- Recipient for academic excellence

SCIENTIFIC ACTIVITIES

C=CONFERENCE, J=JOURNAL, S=SPECIAL SESSION, T=TRACK, W=WORKSHOP

- **[W.21] KNOwledge GRaphs & Agentic Systems Interplay (NORA) at AACL-IJCNLP** 2026
Reviewer 
- **[W.20] RAGE-KG at ISWC** 2026
Reviewer 
- **[T.5] Industry Track at EMNLP** 2026
Reviewer 
- **[W.19] GenAIK-NORA at IJCAI-ECAI** 2026
Reviewer 
- **[W.18] SemEval-2026 Task 11 at SemEval** 2026
Reviewer 
- **[W.17] KG & LLM at LREC** 2026
Reviewer 
- **[W.16] 20th International workshop on Neural-Symbolic Learning and Reasoning (NeSy)** 2026
Reviewer 
- **[W.15] LLM-driven Knowledge Graph and Ontology Engineering (llms4kgoe) workshop at ESWC** 2026
Reviewer 
- **[T.4] Late-Breaking Works and Demos Track at XAI** 2026
Reviewer 
- **[T.3] Industry Track at EACL** 2026
Reviewer 
- **[W.14] NORA workshop at NeurIPS** 2025
Reviewer 
- **[T.2] Industry Track at EMNLP** 2025
Reviewer 
- **[J.2] Engineering Applications of Artificial Intelligence** 2025
Reviewer 
- **[W.13] Assessing World Models: Methods and Metrics for Evaluating Understanding workshop at ICML** 2025
Reviewer 
- **[W.12] 19th International workshop on Neural-Symbolic Learning and Reasoning (NeSy)** 2025
Reviewer 
- **[W.11] Representational Alignment workshop at ICLR** 2025
Reviewer 
- **[J.1] Neurosymbolic Artificial Intelligence** 2025
Reviewer 
- **[C.6] 3rd World Conference on eXplainable Artificial Intelligence (XAI)** 2025
Programme Committee 
- **[T.1] Industry Track at EMNLP** 2024
Reviewer 
- **[W.10] NeurIPS workshop on Behavioral Machine Learning** 2024
Reviewer 
- **[S.1] Harmonising Generative AI and Semantic Web Technologies at ISWC** 2024
Programme Committee 

• [W.9] Large Language Models and Cognition workshop at ICML <i>Reviewer</i>	2024 
• [C.5] 4th International Conference on NLP & Data Mining (NLDM) <i>Programme Committee</i>	2024 
• [C.4] 10th International Conference on Computer Science and Information Technology (CSTY) <i>Programme Committee</i>	2024 
• [C.3] 2nd World Conference on eXplainable Artificial Intelligence (XAI) <i>Programme Committee</i>	2024 
• [C.2] 5th International conference on Advanced Natural Language Processing (AdNLP) <i>Programme Committee</i>	2024 
• [W.8] 18th International workshop on Neural-Symbolic Learning and Reasoning (NeSy) <i>Programme Committee</i>	2024 
• [C.1] 2nd International Conference on NLP & AI (NLP AI) <i>Programme Committee</i>	2024 
• [W.7] Neuro-Symbolic Learning and Reasoning in the Era of Large Language Models workshop at AAAI <i>Reviewer</i>	2024 
• [W.6] 17th International workshop on Neural-Symbolic Learning and Reasoning (NeSy) <i>Programme Committee</i>	2023 
• [W.5] Neurosymbolic Generative Models (NeSy-GeMs) workshop at ICLR <i>Programme Committee</i>	2023 
• [W.4] 5th Financial Narrative Processing (FNP) workshop <i>Programme Committee</i>	2023 
• [W.3] FinCausal shared task at The 4th Financial Narrative Processing (FNP) workshop <i>Organizer</i>	2022 
• [W.2] FinCausal shared task at The 3rd Financial Narrative Processing (FNP) workshop <i>Organizer</i>	2021 
• [W.1] FinCausal shared task at The 2nd Financial Narrative Processing (FNP) workshop <i>Organizer</i>	2020 

TEACHING ACTIVITIES

B=B.S., M=M.Sc.

• [M.14] Intensive Pre-Semester Bootcamp <i>SciencesPo Paris School of International Affairs</i> ◦ Language: English ◦ Number of hours: 6 ◦ Description: The bootcamp aims to refresh essential quantitative reasoning skills and to introduce key concepts in data, algorithms, machine learning and artificial intelligence	2026-Current 
• [B.2] Relational Database Management Systems <i>Data ScienceTech Institute School of Engineering</i> ◦ Language: English ◦ Number of hours: 25 ◦ Description: Installation and system configuration, fundamentals of administration	2026-Current 
• [M.13] Programming, Databases, and Machine Learning Foundations for Decision-Makers <i>SciencesPo Paris School of International Affairs</i> ◦ Language: English ◦ Number of hours: 24 ◦ Description: A practical introduction to computational tools and thinking that covers the basics of programming, data management, and machine learning, equipping students with skills to work with data in modern research and policy contexts	2025-Current 
• [M.12] Computer Architecture <i>Data ScienceTech Institute School of Engineering</i> ◦ Language: English ◦ Number of hours: 5 ◦ Description: Fundamentals of computer architecture & operating systems, from the Turing machine to (not so) modern architectures, fundamentals of hardware architectures from the BIOS/UEFI via the kernel to the shell	2025-Current 
• [M.11] Data Management <i>Data ScienceTech Institute School of Engineering</i> ◦ Language: English ◦ Number of hours: 5 ◦ Description: Fundamentals of information systems analysis & design, functional dependency, relation model & relational algebra	2024-Current 

- **[M.10] Operating Systems and Networking**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 5
 - Description: Fundamentals of algorithmics and data structure design, network layers, routing & protocols, address spaces and associated service with the TCP/IP suite

2024-Current

- **[B.1] Relational Model and SQL**
Data ScienceTech Institute School of Engineering

 - Language: French, English
 - Number of hours: 25
 - Description: Introduction to the relational model, relational algebra operations, normal forms, database system design & SQL queries

2023-Current

- **[M.9] Deep Learning**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 25
 - Description: PyTorch introduction, GPU training, neural network applications (computer vision, natural language processing, reinforcement learning)

2023-2025

- **[M.8] Artificial Neural Networks**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 25
 - Description: Shallow & deep artificial neural networks, data representation & distributed representations, backpropagation & gradient descent, training & optimization, Python applications (Tensorflow & Keras)

2023-2025

- **[M.7] Amazon AWS**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 50
 - Description: Introduction to cloud computing, cloud architecture, preparation for the "AWS Certified Solutions Architect – Associate" certification

2023-2025

- **[M.6] Data Wrangling with SQL**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 25
 - Description: Review of relational models & database management systems, advanced SQL queries, dynamic SQL, stored procedures & triggers

2023-Current

- **[M.5] AI Awareness**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 5
 - Description: Learning processes in living organisms and computers, AI ethics and challenges

2023-Current

- **[M.4] Software Engineering - Part 2**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 25
 - Description: Python object-oriented programming, overview of C++

2021-2025

- **[M.3] Python Machine Learning Labs**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 25
 - Description: Data structures, cleaning & preparation, Python libraries, feature engineering, machine learning algorithms, classification & regression

2021-Current

- **[M.2] Introduction to Data Structures & Software Engineering**
Data ScienceTech Institute School of Engineering

 - Language: English
 - Number of hours: 10

2021-2022


- Description: Introduction to data structures, benchmarks in Python & R, introduction to data simulation
- **[M.1] Clean IT** 2021-Current
Data ScienceTech Institute School of Engineering 
- Language: English
- Number of hours: 5
- Description: Introduction to Google Colab, Python local setup, virtual environments, IDE configuration, R & RStudio, Git & GitHub

LITERARY PUBLICATIONS

A=ANTHOLOGY, B=BOOK, J=JOURNAL

- **[B.10] Dégénérescence/Endgame** 08/2026
Amazon Kindle Direct Publishing 
- **[J.32] Lettre à l'humanité** 2026
Printemps des Poètes Recueil Numéro 14 
- **[B.9] Game Theory** 09/2024
Amazon Kindle Direct Publishing 
- **[A.6] The Poetryhood** 2024
The Poetryhood 
- **[J.31] What's in a Name?** 2024
The Paris Institute For Critical Thinking 
- **[A.5] We Call to the Eye & the Night Love Poems by Writers of Arab Heritage** 2023
Persea Books 
- **[J.30] Winter 2023/24** 2023
The Paradox Literary Magazine 
- **[J.29] Issue 7** 2023
Indelible Lit 
- **[J.28] Volume 1 Issue 3** 2023
The Literatus 
- **[J.27] Issue 4** 2023
Tap into Poetry 
- **[B.8] Titans** 08/2023
Amazon Kindle Direct Publishing 
- **[A.4] Winter** 2022
Poets Choice 
- **[A.3] Impressions of Love** 2022
Pencil 
- **[B.7] Memory** 08/2022
Waterton Publishing Company 
- **[J.26] Featured Writer** 2022
Cajun Mutt Press 
- **[J.25] Spring 2022** 2022
New Note Poetry 
- **[J.24] Issue 4** 2022
The Pine Cone Review 
- **[J.23] Poem** 2022
Rusted Radishes 
- **[J.22] Issue VII** 2022
The Closed Eye Open 
- **[J.21] Issue 1** 2021
BEATIFIC 
- **[J.20] Poem** 2021
Jasad 
- **[J.19] Poem** 2021
Salad Days 
- **[J.18] Poem** 2021
Sledgehammer Lit 
- **[B.6] Correspondence** 08/2021
Waterton Publishing Company 
- **[B.5] Stems from the Edge of Silence** 05/2020
Waterton Publishing Company 
- **[J.17] Clogs** 2020
Jellyfish Review 

• [J.16] Terminal One <i>Unite the World</i>	2020 
• [J.15] Issue 13 <i>Emerge Literary Journal</i>	2020 
• [J.14] Poems <i>Kreaxxxion Review</i>	2020 
• [J.13] Poem <i>Hamline Lit</i>	2020 
• [J.12] Poem <i>Former People</i>	2020 
• [J.11] Issue 1.7 <i>FEED Lit Magazine</i>	2020 
• [J.10] Poem <i>Open Arts Forum</i>	2020 
• [J.9] Volume 4 <i>Press Pause Press</i>	2020 
• [J.8] Poems <i>Spillwords Press</i>	2020 
• [J.7] Issue 34 <i>Poet Speak Magazine</i>	2020 
• [J.6] Poem <i>Poetry Potion</i>	2020 
• [J.5] Poem <i>Oddball Magazine</i>	2020 
• [J.4] Issue 137 <i>Red Fez</i>	2020 
• [J.3] Volume 3 Issue 2 <i>Ink In Thirds</i>	2019 
• [B.4] It's a Big Big World <i>Waterton Publishing Company</i>	07/2019 
• [A.2] And We Chose Everything <i>Turning Point Books</i>	2018 
• [B.3] Vitality <i>Waterton Publishing Company</i>	12/2018 
• [B.2] Diary In Poems <i>G&J Productions</i>	04/2018 
• [B.1] A Road Away From Home <i>G&J Productions</i>	01/2017 
• [A.1] One Hundred Voices: Volume One <i>Allegiant Publishing Group</i>	2016 
• [J.2] Paying Dues <i>Talking Soup</i>	2016 
• [J.1] Volume 2 Issue IX <i>The Rain, Party, & Disaster Society</i>	2015 

VOLUNTEER EXPERIENCE

• First Aid Training <i>Protection Civile Paris Seine</i> ◦ Passed the "PSC1 - Prévention et Secours Civiques de niveau 1" certification	07/2023 - 08/2023
• Mentor <i>JobIRL L'orientation In Real Life</i> ◦ Mentorship and job orientation for students looking to enter the data field	06/2022 - 02/2023
• Mentor <i>Social Builder</i> ◦ Mentor in the Women in Digital (WID) program ◦ Mentor in the Adabot program	09/2021 - 02/2023
• Corporate Social Responsibility Member <i>Yseop</i> ◦ Member of the CSR team ◦ Responsible for motivating local and global company initiatives ◦ Liaison with associations and organizations	03/2021 - 02/2023

• Ambassador Data ScienceTech Institute School of Engineering ◦ Student mentorship and job orientation	01/2020 - 02/2023
• Editor Kiss My Poetry ◦ Content selection and front-page editor	07/2020 - 12/2020
• Volunteer Greater Paris University Hospitals - AP-HP ◦ Data Science volunteer for COVID-19 data analysis tasks	03/2020 - 07/2020
• Ambassador Beirut Poetics ◦ Ambassador of the Beirut Poetics Lebanon branch journal ◦ Local poetry event organization and engagement of community members ◦ Promotion of cultural outreach of poetry	07/2018 - 12/2022

PROFESSIONAL MEMBERSHIPS

• Procedural Memory Knowledge Representation W3C Community Group	2026 - Present
• Web&AI W3C Community Group	2026 - Present
• Global Network on Large-Scale Cross-domain and Multilingual Open Knowledge Graphs	2025 - Present
• Neuro-symbolic AI Interest Group at the Alan Turing Institute	2024 - Present
• Cognitive AI W3C Community Group	2024 - Present
• Artificial Intelligence Knowledge Representation W3C Community Group	2024 - Present
• Knowledge Graph Construction W3C Community Group	2024 - Present
• Knowledge Graphs Interest Group at the Alan Turing Institute	2024 - Present
• GDR RADIA	2024 - Present
• Association for Computational Linguistics	2022 - Present

CERTIFICATIONS

• Wikidata: Wikidata Ontology Course	06/2026
• IBM: Neuro-Symbolic AI Essentials	08/2022
• Neo4j: Certified Professional	07/2022
• Udacity: Machine Learning DevOps Engineer	05/2022
• Hasso Plattner Institute: Knowledge Graphs	03/2022
• Neo4j: Graph Data Science Certified	05/2022
• Massachusetts Institute of Technology: Minds and Machines	02/2021
• Udacity: Deep Reinforcement Learning	01/2021
• Leiden University: Miracles of Human Language: An Introduction to Linguistics	11/2020
• Ecole Polytechnique Federale de Lausanne: Computational Neuroscience	10/2020
• Saint Petersburg State University: Neurolinguistics	10/2020
• The University of Edinburgh: Philosophy and the Sciences	10/2020
• HSE University: Towards language universals through lexical semantics	10/2020
• CentraleSupélec: Analysez vos données textuelles	08/2019
• Université Grenoble Alpes: Introduction à la linguistique de corpus	07/2019
• AWS Academy: Academy Cloud Computing Architecture	12/2018
• Udacity: Artificial Intelligence	07/2017

ADDITIONAL INFORMATION

Languages: English (Native), French (Native), Arabic (Native)

Interests: Literature (Fiction, Philosophy, Classics, Computer Science), Musical Instruments (Electric Guitar), Sports (Football, Basketball, Running, Hiking, Tennis)

Miscellaneous: French driving license

REFERENCES

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